

Reproductive Efficiency and Metabolism of Female Broiler Breeders as Affected by Genotype, Feed Allocation, and Age at Photostimulation. 2. Sexual Maturation

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ABSTRACT A 3 × 4 × 2 factorial design trial was conducted to determine the effect of 3 broiler breeder strains, 4 target BW profiles, and 2 photostimulation ages on sexual maturation traits. A total of 560 pullets from each strain (Hubbard Hi-Y, Ross 508, and Ross 708) were housed in 24 pens. The BW profiles diverged at 4 wk and subsequently converged at 32 wk of age as follows: standard (approximate mean target BW profile of the 3 strains), low (12-wk BW target = 25% lower than standard followed by rapid gain to 32 wk), moderate (12-wk BW target = 150% of standard followed by lower rate of gain to 32 wk), and high (12-wk BW target = 200% of standard followed by minimal growth to 32 wk). Birds were photostimulated at 18 (18WK) or 22 (22WK) wk. Eight birds per interaction (n = 192) were killed and dissected at sexual maturity (first oviposition). Body weight profile affected timing of sexual maturity with the 18WK but not

the 22WK treatment. There was no effect of BW profile on initial egg weight. Mean BW at sexual maturity was 3.40, 3.21, 3.01, and 2.84 kg for high, moderate, standard, and low birds, respectively. This affected shank and keel length, indicating differences in frame size. Ovary weight at sexual maturity was 6 g heavier in the low treatment (55 g) compared with other BW profiles. This was related to recent feeding level, which was highest at sexual maturity in the low treatment. Arrangement of ovarian large yellow follicles (LYF) changed with photostimulation age. Birds that were photostimulated at 18WK had 1.33 LYF hierarchies, compared with 1.22 in 22WK birds. The 18WK-high birds had increased excess LYF production (55% paired) compared with other BW profiles, whereas both small yellow follicle number and atresia were increased. Feeding profiles affected growth and conformational traits, with little effect on reproductive morphology at sexual maturity.

Key words: broiler breeder, sexual maturation, ovarian morphology, growth profile, fleshing

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INTRODUCTION

The process of sexual maturation in the broiler breeder hen represents a major shift in its physiological status. Before sexual maturation, nutrients are deposited in the carcass or used for metabolic functions. Varying pullet feeding programs can affect how nutrients are allocated during sexual maturation—a trait that is affected by genetic strain (Renema and Robinson, 2004). Indicators such as a critical age, BW, and carcass composition, or all, are likely the result of a metabolic signal initiating the cascade of sexual developmental processes (Renema et al., 1999a). Critical age is the point at which the hypothalamo-pituitary-gonadal axis is activated by a photostimulatory cue. The result is the release of gonadotropin-releasing hormone (GnRH) by GnRH neurons. The GnRH stimulates

gonadotropin secretion from the anterior pituitary (Sharp, 1993), which support regulated production of gonadal steroids. Elevated circulating estrogen levels increase hepatic production of yolk lipids destined for the ovary (Walzem, 1996) and consequently on the number of ovarian large yellow follicles (LYF; Renema and Robinson, 2004).

The earliest-maturing birds in a flock are the most likely to have excessive follicle development (Renema et al., 2001a). Providing pullets with excessive nutrients before sexual maturity can vastly increase the rate of yolk deposition (Hocking et al., 1987; Yu et al., 1992). With moderate overfeeding, the number of LYF at sexual maturity can increase to greater than 10 (Hocking et al., 1987; Robinson et al., 1998). Although ad libitum feeding can trigger the formation of over 20 LYF at sexual maturity (Yu et al., 1992), the number of LYF found in highly efficient breeder hens at sexual maturity is in the range of 7 to 8 (Renema and Robinson, 2004).

If most of the hens in a flock are physiologically mature at photostimulation (PS), they will commence lay within a narrow range in time and hence go through production

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as a more uniform group (Robinson et al., 1996). With a high proportion of birds at a common stage of maturity, feed allocation through sexual maturation and early lay will more closely match the requirements of most of the individuals in the flock, contributing to improved flock performance.

This project examined the effects of genetic and environmental (nutritional and photostimulatory) parameters on reproductive performance of female breeders. This manuscript characterizes the effect on the sexual maturation process of 3 strains of hens reared on 4 different BW profiles and photostimulated at 18 or 22 wk of age. Specifically, the role of BW profile in the development of carcass traits and the timing of sexual maturity was determined. The objective was to determine if sexual maturation traits in several genetic strains could be influenced by rearing BW profile during rearing.

MATERIALS AND METHODS

The birds studied in this project were managed according to the Guide to the Care and Use of Experimental Animals (Canadian Council on Animal Care, 1984). The experimental protocol was approved by the University of Alberta's Faculty of Agriculture, Forestry and Home Economics Animal Policy and Welfare Committee.

Experimental Design

The experiment consisted of $3 \times 4 \times 2$ factorial arrangement of treatments with 3 commercial broiler breeder strains, 4 BW profiles from 0 to 32 wk of age, and 2 PS ages. All target BW profiles converged at 3,430 g at 32 wk and were identical after. The 3 strains included in the trial were Hubbard Hi-Y (Hubbard-ISA, Walpole, NH), Ross 508, and Ross 708 (Aviagen Inc., Huntsville, AL). The control BW profile (standard) was a blend of the breeder-recommended target BW profiles for the 3 strains used. A low BW profile was based on an early reduction in feed allocation (12-wk BW was 25% lower than standard), followed by a period of larger feed allocations during sexual maturation. A moderate BW profile was based on a 12-wk BW target of 150% of standard, followed by lower rate of gain to 32 wk. A high profile was based on a 12-wk BW target of 200% of standard, followed by minimal BW gains during the sexual maturation period. A figure of the target BW profiles is presented elsewhere (Robinson et al., 2007). Pullets were photostimulated by an abrupt increase in photoperiod to 15L:9D at either 18 or 22 wk of age. A total of 192 birds were euthanized and dissected on the day following their first oviposition. This population consisted of 64 pullets per strain: 16 per BW profile, 8 photostimulated at 18 wk (18WK), and 8 photostimulated at 22 wk (22WK).

Stocks and Management

Three commercial strains were reared in 2.4×2.9 m floor pens in groups of 70 chicks per pen ($n = 560$ per

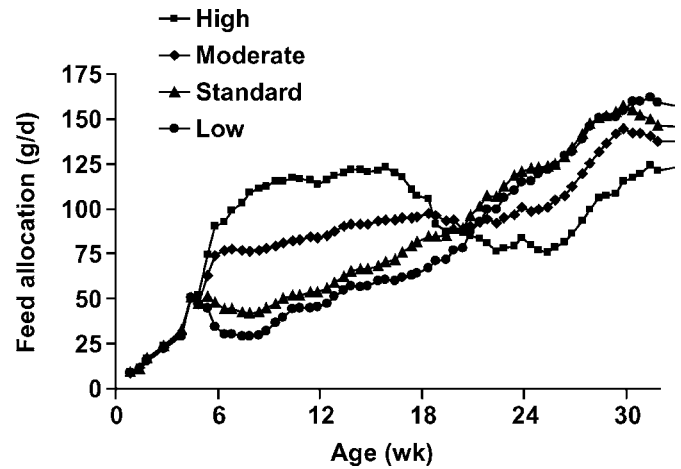


Figure 1. Daily feed allocation for 3 strains of broiler breeder pullets from 0 to 32 wk of age. Four BW curves included standard: mean breeder BW target; low: 12-wk BW target was 25% lower than standard, followed by rapid gain to 32 wk; moderate: 12-wk BW target was 150% of standard, followed by lower rate of gain to 32 wk; and high: 12-wk BW target was 200% of standard, followed by minimal growth to 32 wk.

strain) on straw litter. The chicks were reared following a photoperiod of 24L:0D for the first 3 d and 8L:16D for the remainder of the rearing period. Pullets were randomly preassigned at hatch for analysis of sexual maturity characteristics, evaluation of growth, or reproductive efficiency. Growth and carcass traits at specific times during rearing were assessed by Robinson et al. (2007), and data from birds assessed for reproductive efficiency were assessed by Zuidhof et al. (2007). The current manuscript summarizes bird characteristics at sexual maturity.

A starter diet containing 2,780 kcal/kg and 18% CP was fed from 0 to 5 wk of age; a grower diet containing 2,710 kcal/kg and 15% CP was fed from 5 to 22 wk; and a breeder diet containing 2,830 kcal/kg and 16% CP was fed from 22 wk of age (Robinson et al., 2007). All pullets were fed ad libitum for the first 4 wk, after which daily feed allocations for each pen were determined twice weekly based on BW, recent BW gain, and short-term target BW gain. Feed allocations used to maintain target BW profiles are shown in Figure 1.

Pullets were moved to individual laying cages at 18 wk of age. Feed allocations were based on similar criteria as during the rearing period, with consideration also given to the nutritional demand of egg production. Body weight data were recorded semiweekly until 32 wk of age and weekly thereafter. A common feed allocation was provided for all birds within an interaction (strains $\times$ BW profile $\times$ PS age) except for the period from 18 to 24 wk of age, in which feed was allocated identically to both photostimulation groups.

Data Collection. Sexual maturity (day of first oviposition) was recorded for each bird, and the duration of the interval between photostimulation and first egg was calculated. The weight of the first egg was recorded if it had a normal shell and a single yolk. Each bird was weighed, and shank width (approximately midpoint), right shank length (of the tibiotarsus measured from the

hock to the footpad), and keel length (distance from the hypocleido-clavical joint to the caudal end of the sternum) were measured using a digital calliper (Griffin et al., 2005). Chest width was measured with a digital caliper placed under the wings, 2.5 cm posterior to the cranial end of the keel at the widest point on the chest. Chest girth was measured with a fabric tape at the widest point on the breast, positioned snugly under the wings. The measurement was taken to the nearest 0.5 mm during exhalation.

Breast muscle weight was calculated as the sum of the weights of the pectoralis major and minor. The liver and abdominal fat pad were dissected and weighed. The fat pad included the fat surrounding the gizzard but not the proventriculus. Oviduct weight (emptied of contents) was recorded.

The total ovary was removed and dissected to yield the weights of the LYF (greater than 10 mm). Each LYF was individually weighed. The weight distribution of LYF was evaluated to determine the presence of multiple hierarchies (multiple sets). Such hierarchies were assigned to LYF within 1 g (5 g or greater) or within 0.5 g (less than 5 g) of the LYF nearest in weight. The percentage of the total number of LYF in multiple hierarchies and the total number of hierarchies was calculated. The number of small yellow follicles (SYF; 5- to 10-mm diameter) was recorded. The incidence of atretic SYF and LYF was determined, based on follicle color (lighter colored) and a reduction in turgor pressure (Renema et al., 1999b). A stroma weight value was determined by subtracting the weight of the ovary after the LYF and atretic LYF had been removed from the total ovary weight. The incidence of reproductive abnormalities was recorded, including internal ovulation, internal laying, and ovarian regression.

Chemical Analysis. After dissection, each carcass was individually pressure-cooked for 4 h and homogenized using an industrial blender. Duplicate, representative 100-mL samples of homogenate were collected and freeze-dried. After weighing, a representative subsample of the freeze-dried homogenate was oven-dried. Moisture content was determined from the weights of the freeze-dried homogenate and the oven-dried subsamples. Carcass moisture was calculated from the moisture content of the samples and the original weight of the carcass, correcting for moisture addition and losses during cooking. Body chemical composition was determined as follows: fat by Mojonnier diethyl ether extraction (Mills et al., 1983), CP by measuring N content using a Kjeldhal digest, and ash by combustion in a muffle furnace for 24 h at 550°C (AFLB, 2000).

The homogenized freeze-dried samples were scanned with a Foss NIRSystems 6500 visible-near-infrared (NIR) spectrophotometer (400 to 2,500 nm). The NIR prediction equations were developed from the chemical analysis data, which were then used to estimate chemical composition (CP, crude fat, ash, and moisture) for the entire sample. Because the NIR spectrophotometer scanned a larger sample than that used for chemical analysis, sampling bias was reduced using NIR predictions. Validation in

this laboratory has identified an R^2 value of 0.9486 for the relationship between wet chemistry and NIR for protein content and $R^2 = 0.9819$ for fat content. Further studies of NIR to estimate carcass composition can be found in the literature (Kadim et al., 2005; McDevitt et al., 2005).

Sexual Maturation Profiles. To evaluate the timing of sexual maturity, sexual maturation profiles were estimated using 2 methodologies: 1) ANOVA of age at first egg and 2) a nonlinear egg production model (Yang et al., 1989). The nonlinear model describes age at first egg as a cumulative normal distribution and directly estimates the mean and SD of the age at sexual maturity using the function:

$$\text{maturity} = \frac{100}{1 + \exp^{-\pi/3\sigma^* (\text{age} - \mu)}}$$

where age = age in weeks; Maturity = the percentage of hens within a group that has laid 1 or more eggs; π = the ratio of the circumference of a circle to its diameter (3.14159265); σ = the SD in age at sexual maturity (wk); and μ = the age at sexual maturity, defined as the average age at which half of the birds within a group have laid at least 1 egg. The parameters μ and σ were estimated directly using the NLIN procedure of SAS (SAS 9.1, SAS Institute Inc., Cary, NC).

Statistical Analysis. Sexual maturity data were analyzed as a 3-way ANOVA using the MIXED procedure of SAS (SAS Institute), with BW curve, age at photostimulation, and strain as fixed effects. Differences between least squares means were determined using the diff option, which determines pairwise differences, and were reported as significant at the $P < 0.05$ level.

RESULTS AND DISCUSSION

To 32 wk of age, actual BW profiles of the 4 treatment groups closely resembled the target profiles (Figure 2). By 32 wk, all 4 BW profiles had converged as planned.

Age and BW at Sexual Maturation

Delaying photostimulation from 18 to 22 wk of age delayed age at first egg by 16.4 d (Table 1). Mean interval from lighting to first egg was 41.5 d for the 18WK treatment compared with 29.9 d for 22WK pullets (Table 1). This 14-d lag in the onset of lay is clearly apparent in the comparison of the sexual maturation profiles based on photostimulation age (Figure 3, panel A). This delay in maturation corroborates previous studies (Robinson et al., 1996; Renema et al. 2001b). There was no strain difference in the timing of sexual maturation (Table 1). However, the nonlinear estimation technique (Table 1) and the sexual maturation profile (Figure 3, panel B) suggest that the Ross 708 maturation lagged behind the other 2 strains.

There was a direct relationship between the degree of feed restriction and the length of the delay in onset of lay (Table 1). Pullets on the low BW profile entered lay

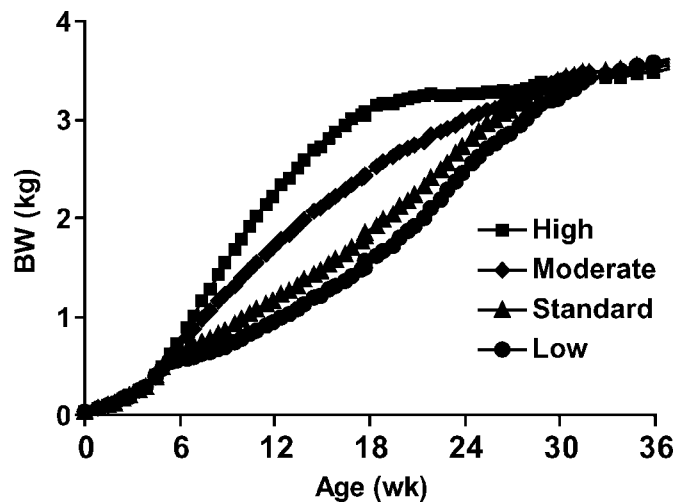


Figure 2. Actual BW profiles for 3 strains of broiler breeder pullets from 0 to 32 wk of age. Four BW curves included standard: mean breeder BW target; low: 12-wk BW target was 25% lower than standard, followed by rapid gain to 32 wk; moderate: 12-wk BW target was 150% of standard, followed by lower rate of gain to 32 wk; and high: 12-wk BW target was 200% of standard, followed by minimal growth to 32 wk.

7 d later than hens on the high BW profile. The sexual maturation profile of the high treatment showed a wide SD in maturation age (Table 1, Figure 3, panel C)—with a subpopulation of high birds entering lay very early, whereas another group was the last to mature. These hens were the most severely feed-restricted during the sexual maturation period as the BW profiles began to converge (Figure 1). The sexual maturation profile of the low treatment started to rise later than for the other groups but exhibited the steepest rise once it commenced (Figure 3, panel C), indicating uniformity. Birds on this profile were fed the most aggressively from 18 wk to sexual maturity (Figure 1). Extra nutrients at this time can shorten the sexual maturation process (Renema et al., 1999b).

A comparison of onset of sexual maturity is made for BW profile within each PS treatment (Table 1). When photostimulation occurred at 18 wk of age, the low hens came into production 17.4 d after the high hens. However, when photostimulation was delayed until 22 wk of age, BW profile did not affect the timing of sexual maturation. For this reason, late (22 wk) photostimulation programs are an equalizer and a great breeder management tool to achieve uniformity in the onset of lay. Improved uniformity of entry into lay associated with delayed lighting is apparent for each strain. In Figures 4 through 6, the uniformity in panel B (22WK) is higher than in panel A (18WK). This supports previous observations that delaying photostimulation can improve the uniformity of sexual maturation, because most birds that are of sufficient BW, fleshing, and hypothalamic maturity to begin lay do not enter production until they receive a photostimulatory cue. At 18 wk of age, fewer birds have reached all the required thresholds to enter lay, which results in a heterogeneous onset of lay response. Such a flock is more difficult to manage, and often results in further

decreased uniformity because of discrepancies in nutrient requirements for egg production.

First Egg Weight

Delaying photostimulation increased the mean weight of the first egg by 4.0 g (Table 1). The 18WK birds started lay 16.4 d before the 22WK birds (Table 1). However, the small early egg size of the 18WK birds persisted, with birds on this treatment laying 31 small eggs (<52 g) compared with 15 eggs in the 22WK birds (data not shown). Small egg production in the current study may have been prolonged by the atypical feed allocation patterns to meet the BW profile targets of the study (Figure 1). The 18WK-standard and 22WK-standard hens laid 23 and 14 small eggs, respectively (data not shown). Neither strain nor rearing BW profile influenced first egg weight (Table 1). Egg size is normally positively correlated with hen size (McDaniel et al., 1981). However, with birds on the heavier BW profiles receiving less feed than those on the lighter BW profiles, this correlation may be neutralized. Feeding level contributed substantially to egg size (Zuidhof et al., 2007).

Carcass Traits at Sexual Maturity

The 237-g greater BW at sexual maturity of 22WK compared with 18WK birds (Table 1) was likely a contributing factor to the greater keel length, chest width, and chest girth measurements in the 22WK hens (Table 2). However, shank traits (length and width) did not differ with PS age. This suggests that assessing adult frame size by chest measurements may be of greater feedback value to broiler breeder managers. Strain did not influence shank length, but Ross 508 birds had the lowest shank width, keel length, chest width, and chest girth. Ross 708 hens had the largest chest width and chest girth, which may be related to increased fleshing rather than increased frame size. The effects of BW profile on frame size were statistically significant for all morphological parameters (Table 2). Shank and keel lengths of the low birds were 10.5 and 13.9 mm, respectively, shorter than that of the high birds. These data are consistent with previous literature (Fatorri et al., 1993).

Hens photostimulated at 22 wk entered lay with significantly more breast muscle mass (absolute basis) than hens photostimulated at 18 wk (Table 2). This may have implications for allocating nutrients to breast muscle maintenance later in lay. As expected, the Ross 708 hens had more breast muscle at first egg compared with Ross 508 and Hubbard Hi-Y hens (21.9 vs. 19.7% of BW, respectively). The effects of BW profile on absolute breast muscle fleshing were directly related to the rearing BW profiles. When expressed as a percentage of BW basis, a linear relationship with level of restriction was not as clear. Birds in the high treatment received substantially less feed around the time of sexual maturation, resulting in very low daily BW gains and a decrease in relative proportion of breast muscle. At sexual maturity, the propor-

Table 1. Age at 50% production and at sexual maturity, maturation interval, maturation BW, and first egg weight of hens of 3 broiler breeder strains reared on varied BW profiles and photostimulated at 18 or 22 wk of age (18WK and 22WK)

Source	Age at 50% production ¹ (d)	Age at sexual maturity ² (d)	Maturation interval ³ (d)	BW at first egg (g)	First egg weight (g)
Photostimulation (PS)					
18WK	167.3 ± 7.8	167.5 ^b	41.5 ^a	2,997 ^b	42.6 ^b
22WK	182.4 ± 6.1	183.9 ^a	29.9 ^b	3,234 ^a	46.6 ^a
SEM	—	1.1	1.1	30	0.6
Strain ⁴					
Hi-Y	174.8 ± 8.2	175.2	35.2	3,109 ^{ab}	44.1
508	173.5 ± 8.8	174.6	34.6	3,057 ^b	44.3
708	178.1 ± 9.0	177.3	37.3	3,180 ^a	45.3
SEM	—	1.3	1.3	37	0.7
BW profile ⁵					
Low	179.6 ± 6.4	179.8 ^a	39.8 ^a	2,839 ^d	44.7
Standard	175.4 ± 7.0	176.5 ^{ab}	36.5 ^{ab}	3,010 ^c	44.9
Moderate	173.3 ± 8.2	173.7 ^b	33.7 ^b	3,213 ^b	44.3
High	172.7 ± 12.8	172.8 ^b	32.8 ^b	3,399 ^a	44.5
SEM	—	1.5	1.5	43	0.9
PS × BW profile					
18WK-low	174.2 ± 5.2	176.8 ^b	50.8 ^a	2,758 ^d	43.7 ^{bc}
18WK-standard	169.5 ± 6.8	169.6 ^c	43.6 ^b	2,846 ^d	42.7 ^{bc}
18WK-moderate	164.1 ± 6.3	164.3 ^{cd}	38.3 ^{bc}	3,094 ^c	42.7 ^{bc}
18WK-high	159.2 ± 7.6	159.4 ^d	33.4 ^{cd}	3,289 ^b	41.3 ^c
22WK-low	184.7 ± 5.0	182.9 ^a	28.9 ^d	2,920 ^d	45.7 ^{ab}
22WK-standard	180.3 ± 5.2	183.5 ^a	29.5 ^d	3,175 ^{bc}	47.0 ^a
22WK-moderate	181.1 ± 5.4	183.0 ^a	29.0 ^d	3,332 ^b	45.9 ^{ab}
22WK-high	184.1 ± 8.6	186.2 ^a	32.2 ^d	3,510 ^a	47.6 ^a
SEM	—	2.1	2.1	60	1.2
Probability					
Source of variation					
PS		<0.0001	<0.0001	0.0004	<0.0001
Strain		0.30	0.30	0.023	0.45
BW profile		0.004	0.004	<0.0001	0.96
PS × strain		0.37	0.37	0.88	0.27
PS × BW profile		<0.0001	<0.0001	0.95	0.32
Strain × BW profile		0.73	0.73	0.95	0.41

^{a-d}With the exception of the age at 50% production column, means within a column and within a source with no common superscript differ significantly ($P < 0.05$). Interaction means are compared within a strain.

¹Mean and SD as estimated directly by nonlinear least squares regression.

²Sexual maturity = mean day of first oviposition (all birds).

³Interval = time between photostimulation and sexual maturity.

⁴Hi-Y = Hubbard Hi-Y (Hubbard ISA, Walpole NH); 508 = Ross 508 (Aviagen Inc., Huntsville, AL); 708 = Ross 708 (Aviagen Inc.).

⁵Four BW profiles included standard: mean breeder BW target; low: 12-wk BW target was 25% lower than standard, followed by rapid gain to 32 wk; moderate: 12-wk BW target was 150% of standard, followed by lower rate of gain to 32 wk; and high: 12-wk BW target was 200% of standard, followed by minimal growth to 32 wk.

tion of breast muscle in the high treatment was greater than in the low treatment but not different from the moderate and standard treatments (Table 2).

Abdominal fat pad weight was greater in late-stimulated hens than it was for 18WK hens (Table 2) on both an absolute and percentage of BW basis. During the 4-wk period from 18 to 22 wk, the nonphotostimulated birds would have allocated excess energy to the fat pad, because there would have been less circulating estrogen, which would have resulted in increased hepatic lipid production. The Ross 708 hens, characterized by the greatest amount of breast muscle fleshing, had the lowest proportion of abdominal fat pad. The treatments of hens that were the most severely feed restricted during rearing had the smallest fat pads. Conversely, liver weights were more correlated with concurrent feed allocation (Table 2). The

lean hens on the low treatment were on average 7 d older at sexual maturity than the fatter hens on the high BW profile. It would appear that fat pad mass at sexual maturity is more of an indicator of long-term nutrient availability, whereas liver mass is more an indicator of short-term nutrient availability.

Carcass Chemical Analyses

The results of carcass chemical analyses are presented in Table 3. The absolute weight data for carcass composition should be discussed with respect to the BW of these 2 groups (18 wk: 2,997 g; 22 wk: 3,234 g). On a percentage of BW basis, late photostimulated pullets entered lay with more carcass lipid but less carcass protein and water and ash than early photostimulated hens. The increased per-

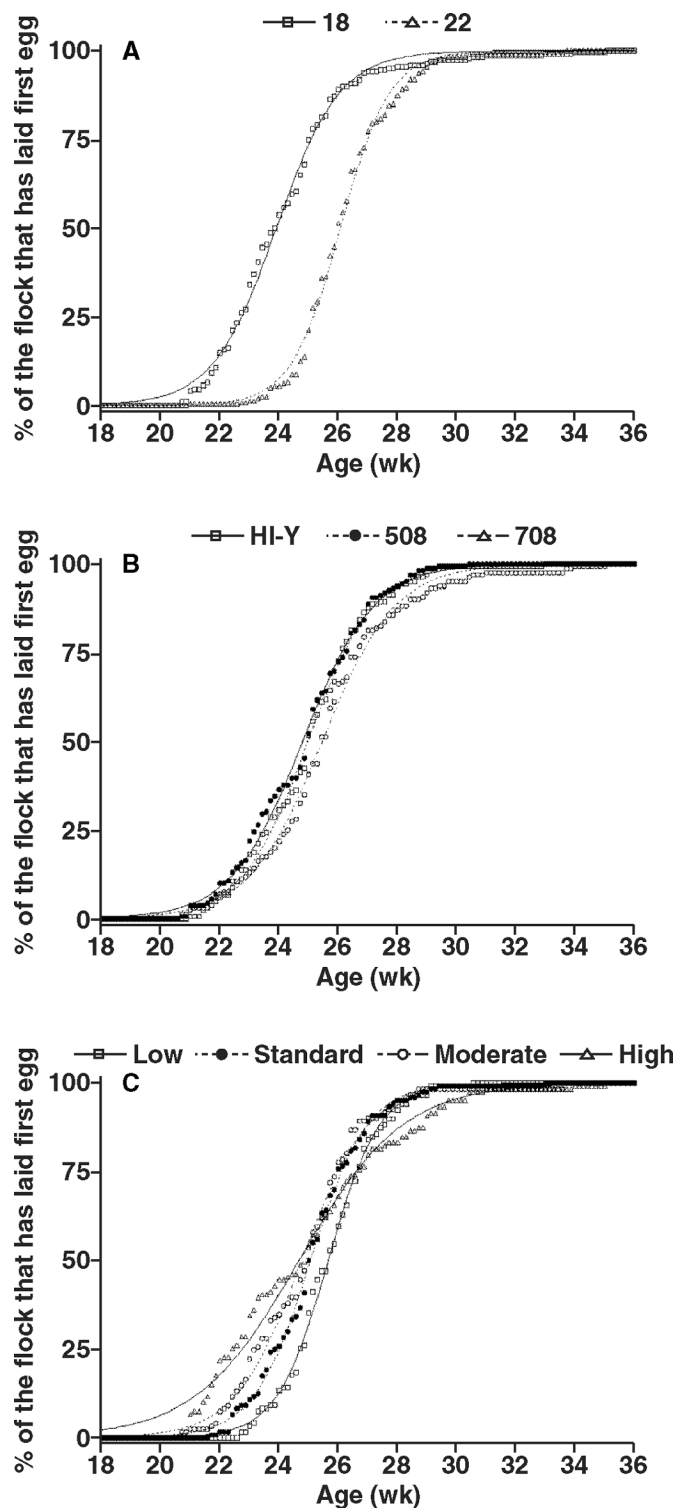


Figure 3. Sexual maturation profile for broiler breeder pullets by photostimulation age (18 or 22 wk of age; panel A), strain (panel B), or pullet BW profile (panel C). Four BW curves included standard: mean breeder BW target; low: 12-wk BW target was 25% lower than standard, followed by rapid gain to 32 wk; moderate: 12-wk BW target was 150% of standard, followed by lower rate of gain to 32 wk; and high: 12-wk BW target was 200% of standard, followed by minimal growth to 32 wk. Hi-Y = Hubbard Hi-Y (Hubbard ISA, Walpole NH); 508 = Ross 508 (Aviagen Inc., Huntsville, AL); 708 = Ross 708 (Aviagen Inc.).

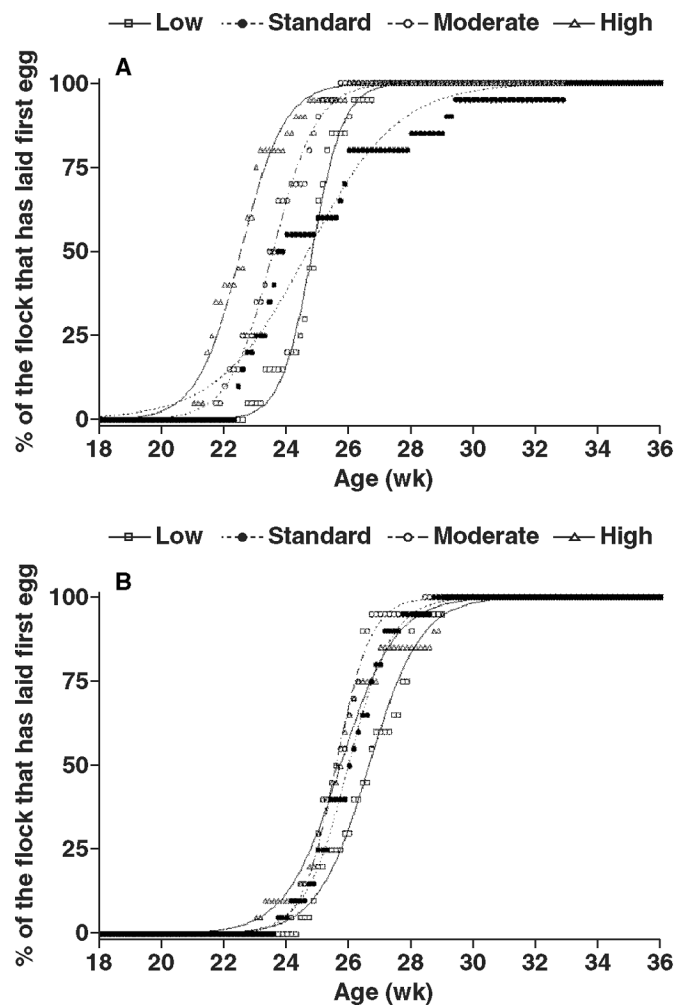


Figure 4. Sexual maturation profile for Hubbard Hi-Y broiler breeder pullets (Hubbard ISA, Walpole NH) from 0 to 56 wk of age, photostimulated at either 18 (panel A) or 22 wk (panel B) of age and reared on 1 of 4 BW profiles: standard: mean breeder BW target; low: 12-wk BW target was 25% lower than standard, followed by rapid gain to 32 wk; moderate: 12-wk BW target was 150% of standard, followed by lower rate of gain to 32 wk; and high: 12-wk BW target was 200% of standard, followed by minimal growth to 32 wk.

centage of carcass ash at first egg in 18-wk photostimulated birds may be related to the difference in BW, in that 22WK birds had a similar frame on which they carried more fat and lean tissue (protein plus water).

Ross 708 and Ross 508 had more carcass protein than the Hubbard Hi-Y hens; conversely, the Hubbard Hi-Y hens had the highest lipid content. Moderate and high hens had a greater proportion of carcass protein and ash than standard and low hens. The percentage of lipid was directly related to the severity of early feed restriction, supporting what was found for abdominal fat pad mass (Table 2).

Reproductive Organ Morphology

Oviduct weight was only influenced by age at sexual maturation (Table 4); 22WK hens had a higher oviduct mass compared with 18WK hens (60.2 vs. 57.6 g, respectively). Oviduct weight at sexual maturity was typically

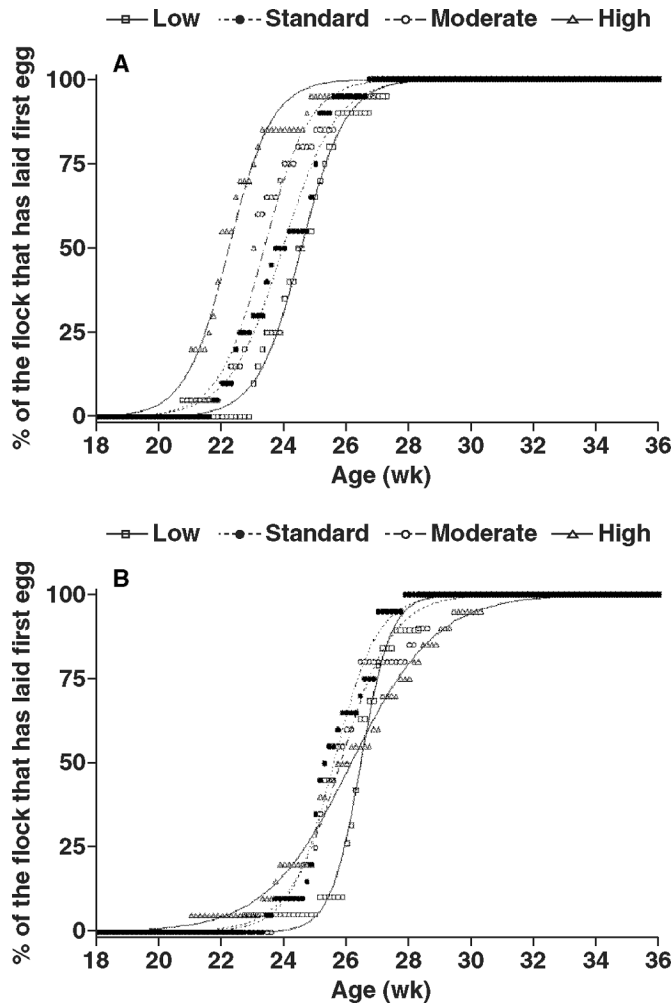


Figure 5. Sexual maturation profile for Ross 508 broiler breeder pullets (Aviagen Inc., Huntsville, AL) from 0 to 56 wk of age, photostimulated at either 18 (panel A) or 22 wk (panel B) of age and reared on 1 of 4 BW profiles: standard: mean breeder BW target; low: 12-wk BW target was 25% lower than standard, followed by rapid gain to 32 wk; moderate: 12-wk BW target was 150% of standard, followed by lower rate of gain to 32 wk; and high: 12-wk BW target was 200% of standard, followed by minimal growth to 32 wk.

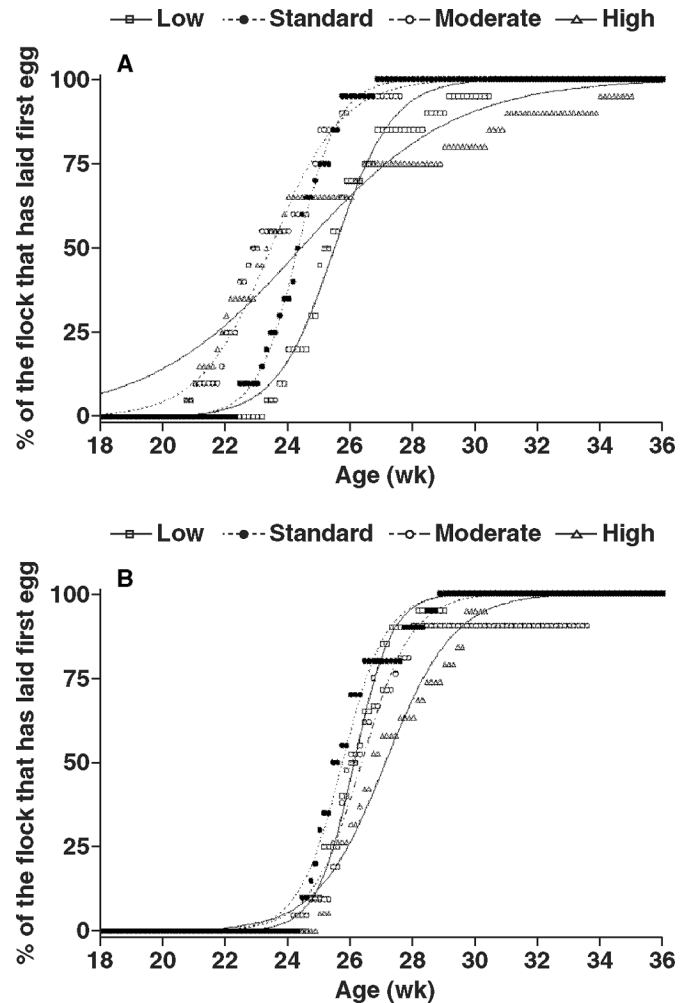


Figure 6. Sexual maturation profile for Ross 708 broiler breeder pullets (Aviagen Inc., Huntsville, AL) from 0 to 56 wk of age, photostimulated at either 18 (panel A) or 22 wk (panel B) of age and reared on 1 of 4 BW profiles: standard: mean breeder BW target; low: 12-wk BW target was 25% lower than standard, followed by rapid gain to 32 wk; moderate: 12-wk BW target was 150% of standard, followed by lower rate of gain to 32 wk; and high: 12-wk BW target was 200% of standard, followed by minimal growth to 32 wk.

from 57 to 60 g and did not vary much outside this range. Ovary weight at first egg was influenced by neither age at PS nor strain. It is hypothesized that the lack of a strain difference in ovary weight was due to all the strains being relatively high breast-yielding strains. Ovary weight was influenced by BW profile. Presumably this effect was related to the level of feed intake during the time of follicle formation rather than the level of restriction during the preceding 16 to 20 wk. The low BW profile hens had the highest feed allocation during sexual maturation and had the heaviest ovaries (55.0 g), which was different from the hens of the other 3 BW profiles (48.4 to 49.4 g).

The ovarian stroma represents the pool of small, undifferentiated follicles. The implications of variation in stroma weight are not well documented; however, high stroma weight likely indicates that a large number of follicles are entering the SYF hierarchy. Hens that were photostimulated at 22 wk had higher stroma weight (Ta-

ble 4). The Ross 708 hens had significantly higher stroma weight than the other 2 strains. Body weight profile did not influence stroma weight; however, substantial variability was apparent in stroma weight.

There were no significant differences in LYF due to the main effects (Table 4). In this trial, all treatment means of the main effects were from 7.27 to 7.81. In studies examining the effects of different rearing profiles and feeding levels near the onset of lay, the conclusion has been that the relationship between BW and LYF numbers is curvilinear (Hocking, 1993). In a subsequent study, this relationship between BW and LYF at the onset of lay was revised to a positive, linear relationship (Hocking, 1996, 2004). This relationship can be modulated by feeding level. Within birds of similar BW, birds consuming more feed will have more LYF (Hocking, 1993). In the current study, ovary weight was significantly higher in low BW profile birds compared with that of any other profile.

Table 2. External carcass measurements and breast muscle, abdominal fat pad, and liver weights at sexual maturation in hens of 3 broiler breeder strains reared on varied BW profiles and photostimulated at 18 or 22 wk of age (18WK and 22WK)

Source	Shank		Keel length (mm)	Chest		Breast muscle weight (% of BW ¹)	Abdominal fat pad weight (% of BW ¹)	Liver weight (% of BW ¹)
	Length (mm)	Width (mm)		Width (cm)	Girth (cm)			
Photostimulation (PS)								
18WK	108.3	11.52	159.5 ^b	96.5 ^b	37.33 ^b	20.7	1.83 ^b	1.58
22WK	108.4	11.65	161.9 ^a	100.1 ^a	37.79 ^a	20.3	2.39 ^a	1.53
SEM	0.4	0.07	0.7	0.7	0.16	0.2	0.07	0.03
Strain ²								
Hi-Y	108.1	11.72 ^a	161.7 ^a	96.9 ^b	37.51 ^{ab}	19.7 ^b	2.28 ^a	1.55
508	108.8	11.41 ^b	159.3 ^b	95.6 ^b	37.19 ^b	19.7 ^b	2.22 ^a	1.54
708	108.3	11.62 ^{ab}	161.1 ^{ab}	102.4 ^a	37.98 ^a	21.9 ^a	1.83 ^b	1.58
SEM	0.5	0.08	0.8	0.8	0.20	0.2	0.08	0.04
BW profile ³								
Low	103.9 ^c	11.10 ^c	153.9 ^d	93.4 ^d	35.90 ^c	19.9 ^c	1.85 ^c	1.78 ^a
Standard	105.4 ^c	11.30 ^c	158.8 ^c	96.4 ^c	36.87 ^b	20.3 ^{bc}	2.04 ^{bc}	1.74 ^a
Moderate	110.4 ^b	11.82 ^b	162.3 ^b	100.2 ^b	38.52 ^a	21.0 ^a	2.19 ^{ab}	1.43 ^b
High	113.9 ^a	12.11 ^a	167.8 ^a	103.3 ^a	38.94 ^a	20.7 ^{ab}	2.36 ^a	1.28 ^c
SEM	0.5	0.10	1.0	1.0	0.23	0.2%	0.10	0.05
Probability								
Source of variation								
PS	0.87	0.18	0.014	0.0002	0.048	0.067	<0.0001	0.31
Strain	0.61	0.028	0.12	<0.0001	0.021	<0.0001	0.0003	0.75
BW profile	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.003	0.002	<0.0001
PS × strain	0.64	0.13	0.024	0.97	0.30	0.27	0.18	0.96
PS × BW profile	0.45	0.34	0.64	0.56	0.18	0.54	0.75	0.28
Strain × BW profile	0.84	0.63	0.75	0.078	0.23	0.39	0.46	0.23

^{a-d}Means within a column and within a source with no common superscript differ significantly ($P < 0.05$).

¹Relative tissue weight.

²Hi-Y = Hubbard Hi-Y (Hubbard ISA, Walpole NH); 508 = Ross 508 (Aviagen Inc., Huntsville, AL); 708 = Ross 708 (Aviagen Inc.).

³Four BW profiles included standard: mean breeder BW target; low: 12-wk BW target was 25% lower than standard, followed by rapid gain to 32 wk; moderate: 12-wk BW target was 150% of standard, followed by lower rate of gain to 32 wk; and high: 12-wk BW target was 200% of standard, followed by minimal growth to 32 wk.

Table 3. Carcass protein, lipid, ash, and water content at sexual maturation in hens of 3 broiler breeder strains reared on varied BW profiles and photostimulated at 18 or 22 wk of age (18WK and 22WK)

Source	Carcass protein		Carcass lipid		Carcass ash		Carcass water	
	(g)	(% of BW)	(g)	(% of BW)	(g)	(% of BW)	(g)	(% of BW)
Photostimulation (PS)								
18WK	636.5 ^b	21.6 ^a	378.5 ^b	12.7 ^b	102.2	3.47 ^a	1,839.1 ^b	62.5 ^a
22WK	672.7 ^a	21.2 ^b	476.2 ^a	14.9 ^a	104.8	3.30 ^b	1,927.0 ^a	60.8 ^b
SEM	6.4	0.1	10.1	0.3	1.2	0.03	18.2	0.2
Strain ¹								
Hi-Y	642.4 ^b	21.0 ^b	466.1 ^a	15.0 ^a	106.5 ^a	3.48 ^a	1,851.6 ^b	60.9 ^b
508	644.0 ^b	21.4 ^a	422.3 ^b	13.9 ^b	102.6 ^{ab}	3.42 ^a	1,840.0 ^b	61.5 ^b
708	677.3 ^a	21.7 ^a	393.8 ^b	12.5 ^c	101.4 ^b	3.25 ^b	1,956.6 ^a	62.7 ^a
SEM	7.8	0.1	12.5	0.3	1.4	0.03	22.5	0.3
BW profile ²								
Low	590.2 ^d	21.1 ^b	350.1 ^d	12.4 ^c	91.5 ^d	3.28 ^b	1,767.3 ^d	63.3 ^a
Standard	621.0 ^c	21.0 ^b	399.7 ^c	13.4 ^{bc}	98.3 ^c	3.34 ^b	1,844.0 ^c	62.5 ^a
Moderate	680.5 ^b	21.7 ^a	444.9 ^b	14.1 ^b	108.0 ^b	3.45 ^a	1,918.0 ^b	61.1 ^b
High	726.6 ^a	21.7 ^a	514.8 ^a	15.3 ^a	116.2 ^a	3.48 ^a	2,002.8 ^a	59.9 ^c
SEM	9.0	0.1	14.2	0.4	1.7	0.04	25.7	0.3
Probability								
Source of variation								
PS	0.12	0.0012	0.0001	0.0001	0.12	0.0001	0.0007	0.0001
Strain	0.034	0.0001	0.0002	0.0001	0.034	0.0001	0.0004	0.0001
BW profile	0.0001	0.0001	0.0001	0.0001	0.0001	0.0003	0.0001	0.0001
PS × strain	0.24	0.35	0.51	0.68	0.24	0.90	0.31	0.91
PS × BW profile	0.36	0.21	0.30	0.68	0.36	0.19	0.41	0.53
Strain × BW profile	0.059	0.91	0.47	0.64	0.059	0.14	0.52	0.77

^{a-d}Means within a column and within a source with no common superscript differ significantly ($P < 0.05$).

¹Hi-Y = Hubbard Hi-Y (Hubbard ISA, Walpole NH); 508 = Ross 508 (Aviagen Inc., Huntsville, AL); 708 = Ross 708 (Aviagen Inc.).

²Four BW profiles included standard: mean breeder BW target; low: 12-wk BW target was 25% lower than standard, followed by rapid gain to 32 wk; moderate: 12-wk BW target was 150% of standard, followed by lower rate of gain to 32 wk; and high: 12-wk BW target was 200% of standard, followed by minimal growth to 32 wk.

Table 4. Oviduct, ovary, and stroma weights; normal and atretic ovarian follicle numbers; F1 weight; and large yellow follicle hierarchy arrangement at sexual maturation in hens of 3 broiler breeder strains reared on varied BW profiles and photostimulated at 18 or 22 wk of age (18WK and 22WK)

Source	Oviduct weight (g)	Ovary weight (g)	Stroma weight ¹	Normal ovarian follicles ²		Atretic ovarian follicles ²		F1 weight ³ (g)	Hierarchies ⁴ (n)
				SYF (n)	LYF (n)	SYF (n)	LYF (n)		
Photostimulation (PS)									
18WK	57.6 ^b	49.3	5.53 ^b	8.06	7.58	0.89	0.438	10.5 ^b	1.33 ^a
22WK	60.2 ^a	51.8	6.30 ^a	9.28	7.27	1.04	0.287	11.9 ^a	1.22 ^b
SEM	0.8	1.3	0.19	0.46	0.19	0.16	0.100	0.3	0.03
Strain ⁵									
Hi-Y	60.4	50.0	5.67 ^b	7.70 ^b	7.63	0.83	0.712 ^a	10.2 ^b	1.36 ^a
508	58.2	49.2	5.67 ^b	8.20 ^b	7.34	0.84	0.172 ^b	11.4 ^a	1.23 ^b
708	58.1	52.4	6.40 ^a	10.12 ^a	7.31	1.21	0.203 ^b	11.9 ^a	1.24 ^b
SEM	1.0	1.6	0.24	0.56	0.24	0.20	0.122	0.3	0.03
BW profile ⁶									
Low	59.0	55.0 ^a	6.09	8.25	7.81	0.90	0.470	12.1 ^a	1.29
Standard	59.3	48.4 ^b	5.74	8.21	7.15	0.79	0.542	10.6 ^b	1.24
Moderate	58.5	49.4 ^b	5.80	8.32	7.42	0.83	0.229	10.9 ^b	1.28
High	58.8	49.4 ^b	6.02	9.92	7.33	1.33	0.208	11.2 ^{ab}	1.30
SEM	1.2	1.9	0.27	0.65	0.27	0.23	0.141	0.4	0.04
PS × BW profile									
18WK-low	58.6	54.1 ^{ab}	6.30 ^{ab}	9.17 ^{abc}	7.83 ^{ab}	1.29 ^{ab}	0.583	11.6 ^a	1.31 ^b
18WK-standard	57.4	43.2 ^c	4.90 ^c	6.75 ^c	6.83 ^{bc}	0.71 ^b	0.583	10.0 ^c	1.25 ^{bc}
18WK-moderate	56.8	49.6 ^{abc}	5.38 ^{bc}	7.54 ^{bc}	7.54 ^{abc}	0.96 ^b	0.375	10.3 ^{bc}	1.31 ^b
18WK-high	57.6	50.3 ^{abc}	5.53 ^{abc}	8.79 ^{abc}	8.13 ^a	0.58 ^b	0.208	10.0 ^c	1.47 ^a
22WK-low	59.4	55.9 ^a	5.88 ^{abc}	7.33 ^{bc}	7.79 ^{ab}	0.50 ^b	0.357	12.6 ^a	1.27 ^{bc}
22WK-standard	61.2	53.5 ^{ab}	6.57 ^a	9.67 ^{ab}	7.46 ^{abc}	0.88 ^b	0.500	11.1 ^{abc}	1.22 ^{bc}
22WK-moderate	60.2	49.2 ^{abc}	6.22 ^{ab}	9.10 ^{abc}	7.30 ^{abc}	0.69 ^b	0.083	11.5 ^{ab}	1.26 ^{bc}
22WK-high	60.1	48.5 ^{bc}	6.52 ^a	11.04 ^a	6.54 ^c	2.08 ^a	0.208	12.4 ^a	1.13 ^c
SEM	1.6	2.6	0.38	0.92	0.38	0.32	0.198	0.5	0.05
Probability									
Source of variation									
PS	0.027	0.18	0.005	0.061	0.25	0.50	0.29	0.0001	0.002
Strain	0.19	0.34	0.042	0.007	0.59	0.29	0.003	0.0005	0.004
BW profile	0.97	0.048	0.76	0.18	0.36	0.28	0.23	0.017	0.68
PS × strain	0.77	0.91	0.76	0.52	0.47	0.67	0.19	0.82	0.39
PS × BW profile	0.83	0.090	0.050	0.049	0.035	0.003	0.88	0.44	0.006
Strain × BW profile	0.34	0.53	0.45	0.80	0.25	0.23	0.73	0.43	0.94

^{a-c}Means within a column and within a source with no common superscript differ significantly ($P < 0.05$).

¹Ovary without the large yellow follicles (follicles > 10 mm in diameter removed).

²SYF = small yellow follicles (5 to 10 mm in diameter); LYF = large yellow follicles (>10 mm in diameter).

³F1 = heaviest large yellow follicle.

⁴Hierarchies = large yellow follicles/groups.

⁵Hi-Y = Hubbard Hi-Y (Hubbard ISA, Walpole NH); 508 = Ross 508 (Aviagen Inc., Huntsville, AL); 708 = Ross 708 (Aviagen Inc.).

⁶Four BW profiles included standard: mean breeder BW target; low: 12-wk BW target was 25% lower than standard, followed by rapid gain to 32 wk; moderate: 12-wk BW target was 150% of standard, followed by lower rate of gain to 32 wk; and high: 12-wk BW target was 200% of standard, followed by minimal growth to 32 wk.

Body weights of the low birds were much less at sexual maturity compared with high birds (2,840 vs. 3,400 g, respectively; Table 1). Increased BW at sexual maturity was not related to increased LYF number. Based on the high ovary weight of low birds, the opposite relationship appeared to have occurred in this study. Like Hocking (1993), we were assessing rearing profile effects and feeding level near maturity. The difference in results between these experiments indicates that current feeding level may be much more important than previously thought in determining the extent of ovary development and LYF numbers compared with the role of BW.

The number of LYF seen in highly efficient breeder hens is in the range of 7.0 to 8.0 (Renema and Robinson, 2004). During sexual maturation, sensitivity to estrogen is high, and surplus nutrients are guided under the influence of estrogen to the liver and ultimately to the ovarian hierarchy. Typically, hens that have less than 7.0 LYF are more prone to problems with persistency than are hens

with more LYF. Hens that have more than 8.0 LYF typically exhibit problems associated with erratic oviposition and defective egg syndrome. Hence, the hens in this study exhibited superior control over follicular recruitment at first egg. There was an important interaction of BW profile and PS age on the numbers of SYF and LYF (Table 4). Standard hens had the fewest SYF when photostimulated at 18 wk, whereas the low hens had the fewest SYF when photostimulated at 22 wk. The physiological significance of having 7 vs. 11 SYF has not been studied well, due to the difficulties in assessing follicle numbers in vivo. Similarly, 18WK-high birds had more LYF than the standard birds. Conversely, however, 22WK-high birds had the fewest LYF, significantly less than the 22WK-low treatment.

The incidence of follicular atresia in SYF and LYF in this trial was low overall in this trial (Table 4). Some differences were seen in the incidence of LYF atresia between strains, with the Hubbard Hi-Y having the highest

incidence (0.712 atretic LYF per bird). There were small but significant differences in the incidence of SYF atresia in means of the interaction of BW profile and age at photostimulation. The highest incidence of this disorder was seen in the high birds that were photostimulated at 22 wk of age. At this time, these birds were receiving the lowest feed allocation, and this may have contributed to this follicle loss. This level of atresia was not significantly different than that observed in the 18WK-low birds. Interestingly, these 2 treatments were the most extreme in terms of a mismatch between PS age and physiological maturity. Hocking and Robertson (2005) reported that strains of breeders that are very heavy (5.6 kg) at sexual maturity have the highest incidence of follicular atresia (18% of all follicles).

On average, every hen had at least 2 unexplained post-ovulatory follicles (data not shown). This means that each hen had 2 follicles ovulated internally, which did not result in the oviposition of normal eggs. None of the treatments influenced the incidence of this phenomenon. This is surprising, because higher rates of unexplained postovulatory follicles are typically seen in birds that are in a positive energy balance (Renema et al., 1999b), which in this trial would be the low birds.

The weight of the largest preovulatory follicle (F1) was greater in 22WK than 18WK hens (Table 4). This is presumably related to increased feed allocation at 22 wk than at 18 wk for most of the hens on the trial. The F1 weight partially explains the observation of increased egg weight in 22WK hens. The Hubbard Hi-Y hens had the lowest F1 weights but had a greater incidence of LYF in multiple sets and a greater number of hierarchies than did any other strain. Hubbard Hi-Y hens may have had poorer control of follicular recruitment at sexual maturity. Such hens would seem to carry a larger compliment of LYF, which would result in a greater likelihood of multiple sets. The low hens had the highest F1 weight (12.1g), seemingly because of current feed intake. Overall, the incidence of multiple follicle hierarchies was low; follicle numbers were generally within a normal range. The early photostimulated hens had 1.33 complete hierarchies, compared with 1.22 in birds photostimulated at 22 wk. When birds were photostimulated at 18 wk, the high birds had the greatest incidence of multiple hierarchies (1.47 sets). From 18 to 22 wk of age, the high pullets were subjected to decreasing feed intake, which resulted in the lowest incidence of multiple hierarchies (1.13). The incidence of multiple hierarchies is a subtle indicator of early stages of erratic oviposition and defective egg syndrome. Renema et al. (2001a) showed that photostimulating at 19 or 21 wk did not influence the incidence of multiple sets. In older genetics, heavier strains have a greater number of LYF than lighter BW strains (Udale et al., 1972; Reddy and Siegel, 1976). In newer genetics, early maturing individuals within a population have larger hierarchies (Hocking et al., 1988). Larger hierarchies are associated with more multiple follicle sets (Hocking, 1992; Renema et al., 1998).

Earlier egg production can be seen with earlier photostimulation; however, this early production is associated with small eggs. Nutrient allocation programs that limit BW gains at the onset of sexual maturity delay sexual development. When pullets were photostimulated early, the BW profiles affected the age at first egg; when birds were photostimulated at 22 wk, BW profile did not influence age at first egg. Feed intake from 18 to 22 wk of age can have a profound effect on carcass growth and reproductive organs. If pullets are subjected to decreasing feed intake during this time, they enter lay with a reduced number of LYF and with less multiple follicles, suggesting that follicular development is closely related to feed intake rather than BW alone. The variation seen in response to genetic strain is less than that seen in response to photostimulation time and BW profile.

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REFERENCES

- AFLB. 2000. Standard Operation Procedures for Agri-Food Laboratories Branch (AFLB), Food Safety Division, Alberta Agriculture, Food and Rural Development, Edmonton, Alberta, Canada.
- Canadian Council on Animal Care. 1984. Guide to the Care and Use of Experimental Animals. Vol. 2. Can. Counc. Anim. Care, Ottawa, Ontario, Canada.
- Fattori, T. R., H. R. Wilson, R. H. Harms, F. B. Mather, R. D. Miles, and G. D. Butcher. 1993. Response of broiler breeder females to feed restriction below recommended levels. 3. Characterizing the onset of sexual maturity. *Poult. Sci.* 72:2044–2051.
- Griffin, A. M., R. A. Renema, F. E. Robinson, and M. J. Zuidhof. 2005. The influence of rearing light period and the use of broiler or broiler breeder diets on 42-day body weight, fleshing and flock uniformity in broiler stocks. *J. Appl. Poult. Res.* 14:204–216.
- Hocking, P. M. 1992. Genetic and environmental control of ovarian function in turkeys at sexual maturity. *Br. Poult. Sci.* 33:437–448.
- Hocking, P. M. 1993. Effects of body weight at sexual maturity and the degree and age of restriction during rearing on the ovarian follicular hierarchy of broiler breeder females. *Br. Poult. Sci.* 34:793–801.
- Hocking, P. M. 1996. The role of body weight and food intake after photostimulation on the ovarian function at first egg in broiler breeder females. *Br. Poult. Sci.* 37:841–851.
- Hocking, P. M. 2004. Roles of body weight and feed intake in ovarian follicular dynamics in broiler breeders at the onset of lay and after a forced molt. *Poult. Sci.* 83:2044–2050.
- Hocking, P. M., A. B. Gilbert, M. Walker, and D. Waddington. 1987. Ovarian follicular structure of White Leghorns fed ad libitum and dwarf and normal broiler breeder pullets fed ad libitum or restricted until point of lay. *Br. Poult. Sci.* 28:495–506.

- Hocking, P. M., A. B. Gilbert, C. C. Whitehead, and M. A. Walker. 1988. Effects of age and early lighting on ovarian function in breeding turkeys. *Br. Poult. Sci.* 29:223–229.
- Hocking, P. M., and G. W. Robertson. 2005. Limited effect of intense genetic selection for broiler traits on ovarian function and follicular sensitivity in broiler breeders at the onset of lay. *Br. Poult. Sci.* 46:354–360.
- Kadim, I. T., O. Mahgoub, and W. Al-Marzooqi. 2005. Prediction of crude protein, extractable fat, calcium and phosphorus contents of broiler chicken carcasses using near-infrared reflectance spectroscopy. *Asian-australas. J. Anim. Sci.* 18:1036–1040.
- McDaniel, G. R., J. Brake, and M. K. Eckmen. 1981. Factors affecting broiler breeder performance. 4. Interrelationship of some reproductive traits. *Poult. Sci.* 60:1792–1797.
- McDevitt, R. M., A. J. Gavin, and S. Andres. 2005. The ability of visible and near infrared reflectance spectroscopy to predict the chemical composition of ground chicken carcasses and to discriminate between carcasses from different genotypes. *J. Near Infrared Spectrosc.* 13:109–117.
- Mills, B. L., F. R. van de Voort, and W. R. Osborne. 1983. Mojonnier method as reference for infrared determination of fat in meat products. *J. Assoc. Offic. Anal. Chem.* 66:1048–1050.
- Reddy, P. R. K., and P. B. Siegel. 1976. Selection for body weight at eight weeks of age. 11. Ovulation and oviposition patterns. *Poult. Sci.* 55:1518–1530.
- Renema, R. A., V. L. Melnychuk, F. E. Robinson, H. L. Classen, and R. D. Crawford. 1998. Reproductive organ morphology and carcass traits in unselected naturally mating female Bronze turkeys at onset of lay. *Can. J. Anim. Sci.* 78:181–187.
- Renema, R. A., and F. E. Robinson. 2003. The impact of varying nutrient allocation from photostimulation on carcass and reproductive traits of conventional and high-yield broiler breeder females. *Poult. Sci.* 82(Suppl. 1):4. (Abstr.)
- Renema, R. A., and F. E. Robinson. 2004. Defining normal: Comparison of feed restriction and full feeding of female broiler breeders. *World's Poult. Sci. J.* 60:511–525.
- Renema, R. A., F. E. Robinson, and P. R. Goerzen. 2001a. Effects of altering growth curve and age at photostimulation in female broiler breeders. 1. Reproductive development. *Can. J. Anim. Sci.* 81:467–476.
- Renema, R. A., F. E. Robinson, P. R. Goerzen, and M. J. Zuidhof. 2001b. Effects of altering growth curve and age at photostimulation in female broiler breeders. 2. Egg production parameters. *Can. J. Anim. Sci.* 81:477–486.
- Renema, R. A., F. E. Robinson, M. Newcombe, and R. I. McKay. 1999a. Effects of body weight and feed allocation during sexual maturation in broiler breeder hens. 1. Growth and carcass characteristics. *Poult. Sci.* 78:619–628.
- Renema, R. A., F. E. Robinson, M. Newcombe, and R. I. McKay. 1999b. Effects of body weight and feed allocation during sexual maturation in broiler breeder hens. 2. Sexual maturation. *Poult. Sci.* 78:629–639.
- Robinson, F. E., R. A. Renema, L. Bouvier, J. J. R. Feddes, M. J. Zuidhof, J. L. Wilson, M. Newcombe, and R. I. McKay. 1998. Effects of photostimulatory lighting and feed allocation in female broiler breeders. 1. Reproductive development. *Can. J. Anim. Sci.* 78:603–613.
- Robinson, F. E., T. A. Wautier, R. T. Hardin, N. A. Robinson, J. L. Wilson, M. Newcombe, and R. I. McKay. 1996. Effects of age at photostimulation on reproductive efficiency and carcass characteristics. 1. Broiler breeder hens. *Can. J. Anim. Sci.* 76:275–282.
- Robinson, F. E., M. J. Zuidhof, and R. A. Renema. 2007. Reproductive efficiency and metabolism of female broiler breeders as affected by genotype, feed allocation and age at photostimulation. 1. Pullet growth and development. *Poult. Sci.* 86:2256–2266.
- Sharp, P. J. 1993. Photoperiodic control of reproduction in the domestic hen. *Poult. Sci.* 72:897–905.
- Udale, R. W., P. B. Siegel, and H. P. Van Krey. 1972. Rates of ovulation and oviposition in growth selected lines of chickens. *Poult. Sci.* 51:2098–2100.
- Walzem, R. L. 1996. Lipoproteins and the laying hen: Form follows function. *Poult. Avian Biol. Rev.* 7:31–64.
- Yang, N., C. Wu, and I. McMillan. 1989. New mathematical model of poultry egg production. *Poult. Sci.* 68:476–481.
- Yu, M. W., F. E. Robinson, R. G. Charles, and R. Weingardt. 1992. Effect of feed allowance for female broiler breeders during rearing and lay. 2. Ovarian morphology and production. *Poult. Sci.* 71:1750–1761.
- Zuidhof, M. J., R. A. Renema, and F. E. Robinson. 2007. Reproductive efficiency and metabolism of female broiler breeders as affected by genotype, feed allocation and age at photostimulation. 3. Reproductive efficiency. *Poult. Sci.* 86:2278–2286.